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2025 Taiwan consensus recommendations for primary care of chronic kidney disease

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ABSTRACT

Chronic kidney disease (CKD) is a major public health challenge in Taiwan, projected to affect over three million people by 2027. Despite national Pay-for-Performance programs, gaps in early detection and risk control call for practical, standardized primary care strategies. To address this, a core group of experts systematically reviewed international and local CKD guidelines and relevant literature, with emphasis on applicability to Taiwan's health system and frontline practice, and drafted preliminary recommendations. A multidisciplinary panel comprising representatives from nephrology, endocrinology, cardiology, family medicine, nursing, nutrition, pharmacy, public health, and government reviewed, discussed, and voted on each recommendation through a formal consensus. Consensus was defined as ≥70 % agreement, with ≥90 % indicating strong consensus. Ten recommendations were endorsed, nine with strong consensus and one achieved consensus. The framework emphasizes risk-based and age-stratified screening, combined eGFR and albuminuria testing for diagnosis, and staging with the G/A classification system. Preventive measures include kidney health literacy promotion and management of modifiable risk factors. In Taiwan, CKD care is embedded within National Health Insurance–supported Pay-for-Performance programs that combine multidisciplinary management to delay disease progression, reduce cardiorenal complications, prolong survival, and improve quality of life. Recommendations also highlight patient self-management, individualized targets for blood pressure, glycemic control, and lipid management, cornerstone therapies with angiotensin-converting enzyme inhibitors/angiotensin II receptor blockers (ACEis/ARBs) and sodium-glucose cotransporter 2 inhibitor (SGLT2i), and timely nephrology referral. The 2025 Taiwan CKD Care Consensus provides evidence-based and locally tailored guidance to strengthen primary care and multidisciplinary collaboration, improving outcomes and sustaining healthcare delivery.

1. Introduction

Chronic kidney disease (CKD) is a major public health challenge in Taiwan, projected to affect over three million people, approximately one in eight adults, by 2027 [1]. It impairs quality of life, drives dialysis demand, and accounts for 36.8 % of catastrophic illness expenditures under National Health Insurance, underscoring its disproportionate economic burden [2–4].

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Although Taiwan continues to have the highest treated end-stage kidney disease (ESKD) prevalence worldwide, the age-standardized incidence of dialysis has gradually declined since 2018 [5], likely reflecting multiple concurrent improvements in CKD care, including early detection and multidisciplinary management promoted through the national CKD Pay-for-Performance (P4P) programs [6,7]. However, suboptimal enrollment limits their population impact. The "Healthy Taiwan 888 Policy" emphasizes control of hypertension, diabetes, and dyslipidemia to reduce chronic disease burden. Real-world evidence suggests that many patients still fail to achieve enrollment, timely intervention, and adequate control [8]. This gap between policy and practice underscores the need for more standardized and integrated approaches.

The 2025 Taiwan CKD Care Consensus was convened to bridge this gap. It provides evidence-based pathways to improve enrollment, timely interventions, and control, aiming to slow CKD progression, reduce dialysis demand, and improve survival and quality of life.

2. Methods

2.1. Evidence review and drafting of recommendations

Five experts reviewed major international (e.g., Kidney Disease: Improving Global Outcomes (KDIGO), American Diabetes Association (ADA)) and local guidelines (e.g., 2025 Taiwan Clinical Practice Guideline for CKD, Early CKD Care Handbook), along with clinical trials, meta-analyses, and systematic reviews. The goal was to distill complex evidence into practical recommendations for primary care. Four core categories—screening and diagnosis, monitoring and referral, risk factor management (hypertension, diabetes, dyslipidemia), and renoprotective therapies—formed the basis of 10 draft recommendations.

2.2. Consensus panel and process

Two preparatory meetings refined the draft before the formal consensus conference. The multidisciplinary panel, composed of 18 members from ten national organizations, included representatives from nephrology, family medicine, endocrinology, cardiology, nutrition, pharmacy, nursing, public health, and the Ministry of Health and Welfare. All were officially nominated by their respective professional societies or agencies. Panelists reviewed and debated each recommendation, with consensus determined by electronic voting: ≥90 % indicated strong consensus, 70–89 % consensus, 50–69 % partial consensus, and <50 % no consensus, subject to revision or omission after discussion.

During the development process, implementation feasibility was a key consideration. The consensus addressed barriers to guideline adoption by clinicians, patients, and health systems [9–13]. To enhance uptake, recommendations were simplified into ten actionable statements organized by care stage and linked with existing workflows. Patient barriers were addressed through kidney health literacy and self-management from early stages, while system barriers were reduced by embedding recommendations within the national P4P Program to ensure policy and reimbursement alignment. Locally relevant thresholds and pathways were adopted for clarity and consistency (Fig. 1).

The panel also evaluated how to represent evidence strength. Formal grading systems such as GRADE were not applied because different components within a single recommendation were supported by varying evidence levels. To maintain practicality, evidence was integrated rather than stratified by hierarchy. Recommendations based mainly on limited or consensus evidence are annotated in the rationale sections for transparency.

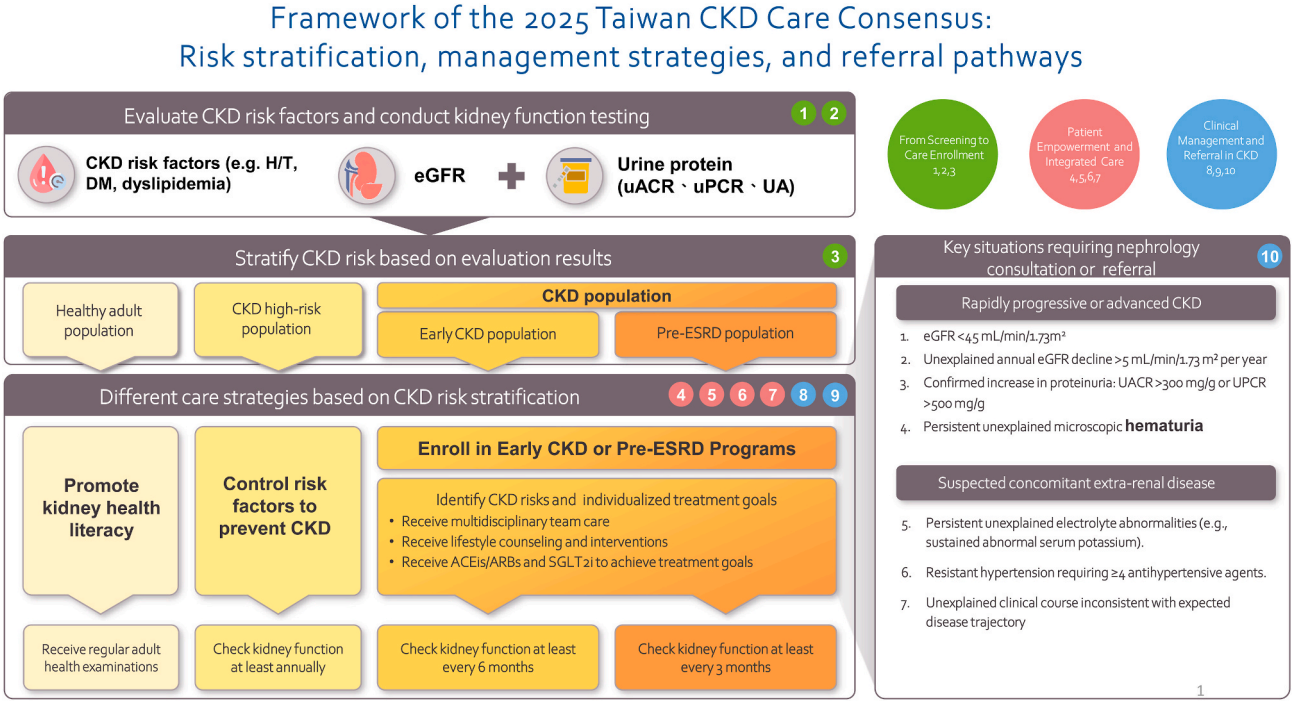


Fig. 1. Framework of the 2025 Taiwan CKD Care Consensus: Risk stratification, management strategies, and referral pathways
Abbreviations list: CKD: Chronic Kidney Disease; H/T: Hypertension; DM: Diabetes Mellitus; eGFR: Estimated Glomerular Filtration Rate; uACR: Urine Albumin-to-Creatinine Ratio; uPCR: Urine Protein-to-Creatinine Ratio; UA: Urinalysis; Early CKD: Early-stage Chronic Kidney Disease; Pre-ESKD: Pre-End-Stage Kidney Disease; ACEis: Angiotensin-Converting Enzyme Inhibitors; ARBs: Angiotensin II Receptor Blockers; SGLT2i: Sodium-Glucose Cotransporter-2 Inhibitor.

3. Results

Tables 1–4 summarize the 10 recommendations and their voting results. Nine achieved strong consensus (≥90 %) and one reached consensus (≥70 %), forming a comprehensive framework for CKD care in Taiwan.

Most recommendations align with international CKD guidelines but are adapted to Taiwan's healthcare context. Recommendations 1–3 (Table 1) cover screening, diagnosis, and staging, incorporating long-term non-steroidal anti-inflammatory drugs (NSAID) use as a locally relevant risk factor and linking screening to the National Health Examination Program. Recommendations 4–5 (Table 2) emphasize kidney health literacy and enrollment in CKD P4P programs for coordinated multidisciplinary care. Recommendations 6–8 (Table 3) focus on patient engagement, lifestyle modification, and individualized targets for blood pressure, glycemia, and lipid control. Recommendations 9–10 (Table 4) address pharmacotherapy and referral criteria, supporting flexible SGLT2 inhibitor use across CKD stages and an earlier referral threshold (eGFR <45 mL/min/1.73 m^2) to promote co-care with nephrologists. Collectively, these recommendations establish an integrated, locally tailored framework for CKD care in Taiwan.

Table 1
CKD definition, screening, and staging (recommendations 01–03).

No.	Recommendation	Level of agreement (%)	Strength of Consensus
01	Adults should undergo CKD risk factor evaluation. Risk factors such as hypertension, diabetes, dyslipidemia, obesity, cardiovascular disease, smoking, history of acute kidney injury, or long-term use of non-steroidal anti-inflammatory drugs (NSAIDs) should be considered. ● For adults with any CKD risk factors: Annual kidney function screening is recommended. ● For adults without CKD risk factors: Kidney function screening according to age may be considered (30–39 years: at least every 5 years; 40–64 years: at least every 3 years; ≥65 years: at least annually).	100	Strong consensus
02	Kidney function screening should include both estimated glomerular filtration rate (eGFR) and proteinuria testing [albuminuria (uACR), total proteinuria (uPCR), or urine dipstick]. CKD diagnosis requires persistence for more than 3 months of any of the following conditions: eGFR <60 mL/min/1.73 m^2 Evidence of abnormal proteinuria (uACR ≥30 mg/g, uPCR ≥150 mg/g, or positive urine dipstick) or hematuria Known structural kidney abnormalities	100	Strong consensus
03	CKD patients should be staged according to eGFR and uACR to stratify the risk of disease progression.	100	Strong consensus

Summary of consensus recommendations on CKD definition, high-risk screening, and staging using combined eGFR and albuminuria.
Recommendations 1–3 were developed based on the KDIGO 2024 CKD Guideline, the 2025 Taiwan CKD Clinical Practice Guideline, and the National Health Examination Program protocol.
Abbreviations list: CKD: Chronic Kidney Disease; eGFR: Estimated Glomerular Filtration Rate; uACR: Urine Albumin-to-Creatinine Ratio; uPCR: Urine Protein-to-Creatinine Ratio.

3.1. Recommendation 01 - Rationale

This recommendation aims to identify adults with or at risk of CKD for timely evaluation in primary care. The 2022 Taiwan Hypertension Guideline advises kidney evaluation in patients with hypertension, while the KDIGO 2022 Diabetes Guideline recommends annual testing in those with diabetes [14,15]. The 2025 Taiwan Clinical Practice Guideline for CKD further identifies hypertension, diabetes, dyslipidemia, obesity, cardiovascular disease, smoking, prior acute kidney injury, and long-term NSAID use as key determinants [16–22]. The KDIGO Controversies Conference has issued similar recommendations [23].

Based on this, adults with ≥1 risk factor should undergo annual testing. For individuals without CKD risk factors, screening intervals may follow the National Health Examination Program: every 5 years for those aged 30–39, every 3 years for those aged 40–64, and annually for those aged 65 or older. Individuals with urinary tract abnormalities, autoimmune disease, or a family history of kidney disease may also be appropriate [23]. This risk factor– and age-stratified framework provides a pragmatic approach for timely CKD identification in primary care. Although evidence on cost-effectiveness remains limited, aligning early detection with the National Health Examination Program supports the consensus-based feasibility in Taiwan.

3.2. Recommendation 02 - Rationale

This recommendation emphasizes establishing a standardized framework for diagnosing CKD by combining kidney function and urine testing, ensuring accurate identification even in early or asymptomatic stages. Early detection is emphasized across multiple guidelines, including the 2025 Taiwan Clinical Practice Guideline for CKD, Early CKD Care Handbook, and KDIGO 2024 CKD Guideline [21,24,25]. All highlight that combining eGFR and urinary protein assessment improves diagnostic accuracy compared with either measure alone. Albuminuria measured by uACR is the preferred marker of glomerular injury, whereas urine protein-to-creatinine ratio (uPCR) and dipstick tests remain practical alternatives. Persistently reduced eGFR or abnormal proteinuria for more than three months establishes the diagnosis of CKD.

KDIGO 2024 CKD Guideline specifies diagnostic criteria when one or more of the following persist ≥3 months: eGFR <60 mL/min/1.73 m^2, abnormal proteinuria or hematuria, structural abnormalities, pathological findings, or relevant medical history indicating cause or contribution to CKD [24]. These criteria enable early identification, even in asymptomatic patients [25].

This consensus emphasizes practicality in primary care by endorsing uPCR and dipstick as feasible alternatives and simplifying diagnostic criteria. Primary care physicians should remain alert, as rare conditions such as tubular or hereditary kidney disorders may go undetected; referral is advised when clinical suspicion persists.

3.3. Recommendation 03 - Rationale

It aims to stratify CKD progression risk using a standardized staging system that combines eGFR and albuminuria. KDIGO 2024 CKD Guideline recommends a two-dimensional framework: eGFR categories (G1 ≥90, G2 60–89, G3a 45–59, G3b 30–44, G4 15–29, G5 <15 mL/min/1.73 m^2) and albuminuria categories (A1 <30, A2 30–300, A3 >300 mg/g) [24] (see Fig. 2). Both reduced eGFR and elevated albuminuria independently and synergistically increase risks of cardiovascular events, hospitalization, kidney failure, and mortality [24]. This combined G/A classification provides prognostic information to guide monitoring frequency and timely interventions based on risk. In addition, primary care physicians are encouraged to use this staging system as a communication tool to enhance patient understanding and support kidney health literacy.

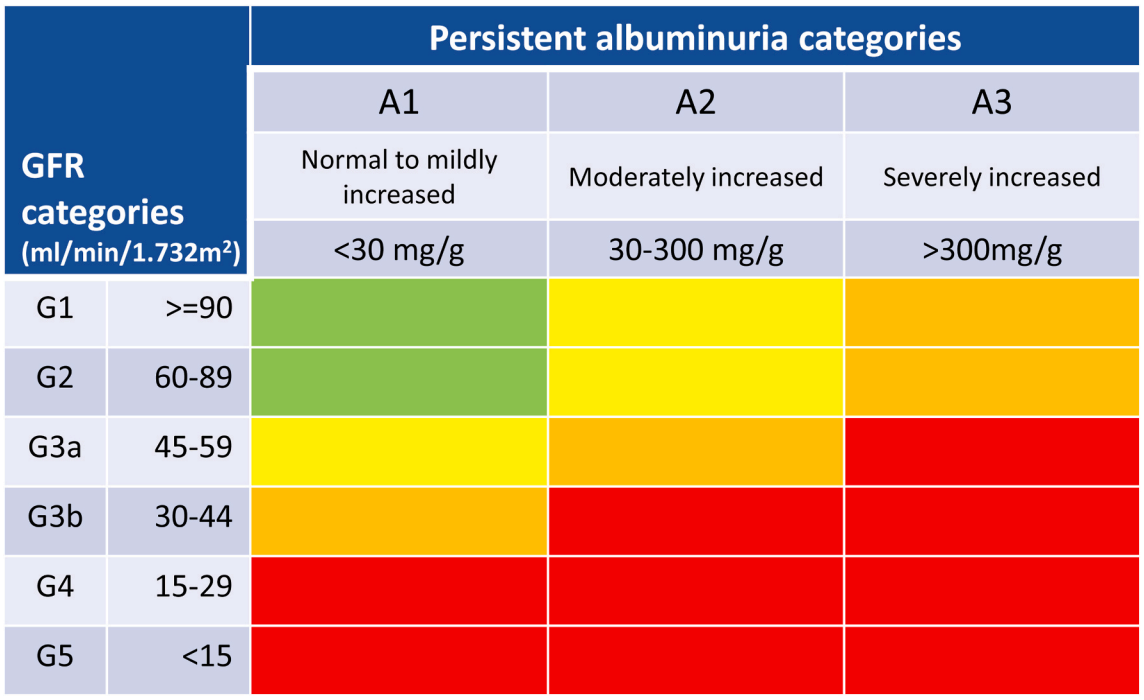


Fig. 2. Risk classification of chronic kidney disease by eGFR and albuminuria
Modified from KDIGO 2012 Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease.
Green: Low risk; Yellow: Moderately increased risk; Orange: High risk; Red: Very high risk
Abbreviations list: GFR: Glomerular Filtration Rate.

Table 2
Kidney health literacy promotion, risk-factor management, and early structured care (recommendations 04–05).

No.	Recommendation	Level of agreement (%)	Strength of Consensus
04	For individuals who do not meet the diagnostic criteria for CKD, kidney health literacy should be strengthened (e.g., adopting regular exercise, maintaining a healthy diet, and avoiding unverified medications), and CKD risk factors should be managed (e.g., smoking cessation, blood pressure, lipid and glucose control, and weight reduction) to prevent the development of CKD.	100	Strong consensus
05	All CKD patients should be enrolled in integrated care programs under Taiwan's Pay-for-Performance initiative (commonly referred to as the Diabetes and Early CKD Care Program and the Pre-ESKD Program) to receive clinical management and patient education through multidisciplinary team care.	100	Strong consensus

Summary of consensus recommendations on universal kidney health literacy promotion for the general population, risk-factor identification and management for individuals at increased CKD risk, and timely enrollment of newly diagnosed CKD patients into structured multidisciplinary care under the national Pay-for-Performance programs (Early-CKD and Pre-ESKD).
Recommendations 4–5 were developed with reference to the KDIGO 2024 CKD Guideline, the 2025 Taiwan CKD Clinical Practice Guideline, and the Taiwan CKD Pay-for-Performance Program, encompassing the Early-CKD and Pre-ESKD programs.
Abbreviations list: CKD: Chronic Kidney Disease; ESKD: End Stage Kidney Disease; Early CKD: Early-stage Chronic Kidney Disease; Pre-ESKD: Pre-End-Stage Kidney Disease.

3.4. Recommendation 04 - Rationale

This recommendation aims to prevent CKD onset in the general population, including those with and without risk factors. Improving kidney health literacy is essential, as low kidney health literacy is associated with the risk of CKD onset, whereas higher literacy improves adherence to risk-factor management, reducing CKD incidence and cardiovascular burden [26].

Lifestyle-related risks include inactivity, unhealthy diet, obesity, hypertension, and diabetes. The KDIGO Controversies Conference emphasized prevention through lifestyle modification and risk factor control [23]. Evidence shows that a balanced diet, regular activity, nonsmoking, adequate sleep, weight management, and control of blood pressure, glucose, and lipids substantially lower CKD and cardiovascular risks [23,27]. The American Heart Association's Life's Essential 8 offers a practical framework for lifestyle modification, encompassing diet, physical activity, sleep, smoking cessation, body weight, blood pressure, cholesterol, and glucose control [28,29]. In overweight patients with diabetes, weight loss has been shown to reduce or delay CKD onset [30].

In addition, social determinants of health (SDOH)—including economic, social, environmental, and psychosocial factors—play a critical role in shaping disparities in CKD, diabetes, and hypertension outcomes [23,28,31,32]. A comprehensive prevention strategy should therefore integrate efforts to enhance kidney health literacy, promote lifestyle modification, and address SDOH across diverse risk groups.

3.5. Recommendation 05 - Rationale

This recommendation emphasizes reducing CKD's public health and economic burden by promoting enrollment in national CKD P4P initiatives—namely, the Diabetes and Early CKD Care Program and the Pre-ESKD Program—which provide structured multidisciplinary care and education [33,34].

The Early CKD Care Program enrolls patients with stages 1–3a (eGFR $\geq$45), while the Pre-ESKD program targets those with severe proteinuria

or stages 3b–5 not yet on dialysis. Robust evidence demonstrates that both programs improve renal outcomes, slow disease progression, lower mortality, and decrease healthcare costs [6,7,35,36]. These results establish integrated care as central to national CKD management.

These programs are delivered by multidisciplinary teams—including nephrologists, primary care physicians, educators, dietitians, pharmacists, and case managers—who collaboratively manage blood pressure, glycemia, lipids, anemia, mineral bone disorder, lifestyle, and preparation for renal replacement therapy. Incorporating decision support systems further enhances care quality, while structured patient education improves engagement and promotes sustained self-management across CKD stages [37]. Collectively, these findings support enrollment in P4P programs as a cornerstone approach to providing comprehensive care, delaying disease progression, and improving patient outcomes.

Table 3
Lifestyle modification, patient engagement, and therapeutic targets in established CKD (recommendations 06–08).

No.	Recommendation	Level of agreement (%)	Strength of Consensus
06	All CKD patients should be informed of the goals of kidney disease management through multidisciplinary team care, which includes: 1 Reducing the risk of mortality. 2 Slowing CKD progression and delaying the need for dialysis. 3 Reducing the occurrence of cardio-renal complications and improving quality of life.	100	Strong consensus
07	The care plan for CKD patients should promote self-management behaviors, such as smoking cessation, healthy diet, regular exercise, weight control, medication adherence, and avoidance of long-term use of non-steroidal anti-inflammatory drugs (NSAIDs) as well as unverified medications.	100	Strong consensus
08	It is recommended that CKD patients should have individualized targets for blood pressure, glycemic control, and lipid management according to CKD stage and clinical status through multidisciplinary team care. **Blood pressure control target:** SBP <130 mmHg, DBP <80 mmHg; in selected patients, SBP <120 mmHg may be considered depending on age, comorbidities, and individual tolerance. **Glycemic control target**: hemoglobin A1c (HbA1c) < 7 %; individualized targets of HbA1c 6.5 %–8.0 % may be set depending on age, comorbidities, and hypoglycemia risk. **Low-density lipoprotein cholesterol (LDL) treatment targets**: ● For CKD patients: LDL <100 mg/dL ● For CKD with diabetes or atherosclerotic cardiovascular disease (ASCVD): LDL <70 mg/dL ● For CKD with both diabetes and ASCVD: LDL <55 mg/d	100	Strong consensus

Summary of consensus recommendations for patients with established CKD, emphasizing lifestyle modification, patient engagement, and individualized therapeutic targets for blood pressure, glycemia, and lipids to optimize outcomes.
Recommendations 6–8 were developed with reference to the KDIGO 2024 CKD Guideline, the ADA 2023 Standards of Care, and the AHA/ACC 2022 Cardiorenal Guidelines, adapted to Taiwan's team-based, Pay-for-Performance CKD care framework.

3.6. Recommendation 06 - Rationale

This recommendation aims to reduce mortality, slow CKD progression, and minimize cardio-renal complications while improving quality of life. These priorities should be clearly communicated to patients through multidisciplinary care. Clear communication ensures patient engagement and adherence while aligning management priorities within the care team.

The KDIGO 2024 CKD Guideline emphasizes a patient-centered strategy integrating education, shared decision-making, and team-based care [24]. Central priorities are reducing progression risk—closely linked to kidney failure, hospitalization, and premature death—and minimizing cardio-renal complications. CKD outcomes include not only kidney failure but also morbidity and mortality from cardiovascular events, infections, and other systemic conditions [38]. These interrelated risks highlight CKD's complexity and the importance of early goal setting to empower patients and coordinate care.

Effective, literacy-adapted communication supported by physicians, nurses, dietitians, and educators—using structured materials, visual aids, and regular follow-up—enhances understanding, empowers patients, and aligns the care team toward optimal outcomes. In addition to clinical outcomes, patient-centered CKD care should incorporate quality-of-life measures, patient preferences, and equitable access to multidisciplinary support. Although formal SDM in Taiwan applies mainly to specific treatment choices, its principles—two-way communication and value alignment—are integrated into ongoing education and follow-up to ensure person-centered care. Applying SDM principles also helps identify and address barriers such as distance, cost, or limited understanding, thereby complementing system-level efforts to improve equitable access to multidisciplinary CKD services.

3.7. Recommendation 07 - Rationale

This recommendation promotes self-care in CKD, emphasizing lifestyle modification alongside pharmacotherapy to slow disease progression, reduce cardiovascular complications, and support active patient self-management.

The KDIGO 2024 CKD Guideline and 2025 Taiwan Clinical Practice Guideline for CKD concordantly endorse a holistic model of care that integrates lifestyle into risk stratification and treatment planning [21,24]. Smoking is a major risk factor for incident CKD and increases vascular events and mortality in CKD and diabetes [39–41]. A healthy diet is associated with lower mortality risk among patients with CKD [42], while regular physical activity reduces the likelihood of developing CKD [41]. Obesity also increases CKD risk [43]. Adherence to prescribed drugs, avoidance of long-term NSAID use, and caution with unregulated supplements are essential for kidney protection [22]. Though evidence is limited, major CKD guidelines like KDIGO, NICE (National Institute for Health and Care Excellence), and KDOQI (Kidney Disease Outcomes Quality Initiative) advise avoiding long-term NSAID use to prevent kidney damage [24,44,45]. These modifiable factors drive CKD incidence, progression, and cardiovascular risk, making patient engagement and behavioral change central to improving outcomes.

In summary, CKD management should go beyond medication to address lifestyle and behavioral factors. Guideline-based multidisciplinary care—delivered through education, counseling, and monitoring—promotes sustained self-management and leads to better renal and cardiovascular outcomes with lower CKD burden.

3.8. Recommendation 08 - Rationale

This recommendation emphasizes the importance of individualized targets for blood pressure (BP), glycemic control, and lipid management in CKD patients. Uniform targets may not be appropriate for all patients,

as CKD severity, age, comorbidities, frailty, and tolerance to therapy can greatly influence both the benefits and risks of intensive control. Individualization ensures that treatment goals maximize cardiovascular and renal protection while minimizing the risks associated with overly aggressive control. While general therapeutic targets are supported by high-level evidence from major randomized trials, individualized goal setting relies on expert consensus to balance benefits and risks in diverse CKD populations.

Bp targets in CKD: Most domestic and international guidelines recommend a target of <130/80 mmHg for patients with CKD and hypertension or albuminuria [25]. The 2022 Taiwan Hypertension Guideline, the 2025 American Heart Association Hypertension Guideline, the KDIGO 2024 CKD Guideline, and the 2025 Taiwan Clinical Practice Guideline for CKD all endorse this threshold [15,21,24,46]. For selected patients who are younger, at lower risk, and able to tolerate intensive therapy, KDIGO 2024 CKD Guideline and the 2025 Taiwan Clinical Practice Guideline for CKD suggest aiming for SBP <120 mmHg to further reduce cardiovascular risk. In all cases, targets should be individualized according to age, comorbidities, baseline risk, and treatment tolerance.

Glycemic targets in CKD: The 2025 Taiwan Clinical Practice Guideline for CKD and Early CKD Care Handbook recommend HbA1c near 7 % without going lower [21,25]. KDIGO 2022 Clinical Practice Guideline for Diabetes Management in CKD, ADA 2025, and the 2024 Taiwan DKD guideline also emphasize individualized targets, generally 6.5 %–8.0 % [14,47,48]. These recommendations underscore the need to individualize glycemic goals, balancing the long-term benefits of intensive control with the risks of hypoglycemia, while considering age, diabetes duration, comorbidities, hypoglycemia risk, and life expectancy.

Lipid target in CKD: The 2025 Taiwan Clinical Practice Guideline for CKD and Early CKD Care Handbook recommend low-density lipoprotein cholesterol (LDL-C) targets of <100 mg/dL for CKD and <70 mg/dL for CKD with diabetes [25,47]. For CKD with both diabetes and atherosclerotic cardiovascular disease (ASCVD), the target is < 55 mg/dL. The 2025 Taiwan Clinical Practice Guideline for CKD and the 2025 Lipid Management Pathway adopt similar thresholds, recommending <100 mg/dL for CKD, <70 mg/dL for CKD with diabetes or ASCVD, and <55 mg/dL for CKD with both conditions [21,49,50]. These guidelines also emphasize early statin initiation and the addition of non-statin agents if needed.

Achieving these individualized targets requires close collaboration among nephrologists, primary care physicians, cardiologists, endocrinologists, dietitians, and nursing staff, underscoring the importance of multidisciplinary care.

3.9. Recommendation 09 - Rationale

This recommendation aims to highlight ACEis/ARBs and SGLT2i as cornerstone therapies for renoprotection in CKD and encourage their use in eligible CKD patients.

ACEis/ARBs: According to the KDIGO 2024 CKD Guideline and the 2025 ADA Standards of Care in Diabetes, renin–angiotensin system inhibitors (RASi), including ACEis and ARBs, should be initiated and titrated to the maximally tolerated dose in patients with CKD and proteinuria [24,51]. Initiation is strongly recommended for CKD patients with severely increased albuminuria and those with diabetes and moderately increased albuminuria, as these groups derive the clearest renal and cardiovascular benefits. In CKD patients with moderately increased albuminuria without diabetes, ACEis/ARBs use is considered appropriate, although the supporting evidence is less robust. Following initiation or dose adjustment of ACEis/ARBs therapy, serum creatinine and potassium should be monitored within 2–4 weeks; if potassium is normal and eGFR decline is <30 %, dosing should be maintained or up-titrated [24].

SGLT2i: SGLT2i are now established as a foundational therapy for CKD patients at risk of progression. Robust evidence demonstrates that they slow eGFR decline, reduce the risk of ESKD, lower hospitalization for heart failure, and improve survival [52,53]. These benefits extend beyond glycemic control and are consistently observed in both diabetic and non-diabetic CKD, particularly in the presence of albuminuria or heart failure. Mechanisms include reduction of glomerular hyperfiltration and proteinuria via tubuloglomerular feedback, lowering tubular workload and oxidative stress, improving glycemic control in diabetes, and promoting natriuresis and improved cardiac energy metabolism in heart failure [54–57].

Recent guidelines converge in recommending SGLT2i as foundational therapy in CKD. Indications consistently include: (1) type 2 diabetes with CKD, (2) CKD with albuminuria (commonly UACR ≥200 mg/g), and (3) heart failure, irrespective of diabetes status. Regarding eGFR thresholds, KDIGO 2024 CKD Guideline and ADA 2025 recommend initiation when eGFR is ≥ 20 mL/min/1.73 m^2, while the 2025 Taiwan Clinical Practice Guideline for CKD specifies higher thresholds. Across these guidelines, therapy is generally continued even as eGFR declines below these thresholds, until dialysis or transplantation, unless contraindicated [21,24,51]. Unlike other guidelines, this consensus does not specify an eGFR threshold for SGLT2i initiation, because major RCTs only enrolled patients with eGFR ≥20 mL/min/1.73 m^2, while emerging real-world data, including studies from Taiwan, suggest potential benefits may extend even to those with lower eGFR [58].

Together, ACEis/ARBs and SGLT2i provide complementary benefits

Table 4
Pharmacotherapy and referral (recommendations 09–10).

No.	Recommendation	Level of agreement (%)	Strength of Consensus
09	It is recommended that CKD patients meeting the following conditions should be treated with the maximally tolerated dose of ACEis/ARBs and SGLT2i: ACEis/ARBs: For CKD patients with proteinuria SGLT2i: For CKD patients with type 2 diabetes, or uACR ≥200 mg/g, or heart failure	88	Achieved consensus
10	Patients with any of the following conditions should consult or be referred to a nephrologist: 1 eGFR <45 mL/min/1.73 m^2 2 Annual decline in eGFR >5 mL/min/1.73 m^2 3 Persistent severe proteinuria (uACR >300 mg/g or uPCR >500 mg/g) 4 Persistent unexplained hematuria 5 Persistent unexplained electrolyte abnormalities (e.g., abnormal serum potassium) 6 Resistant hypertension, requiring at least four or more antihypertensive agents 7 Other unexplained clinical courses inconsistent with the expected disease trajectory	100	Strong consensus

Summary of consensus recommendations on pharmacologic management of CKD, emphasizing ACEIs/ARBs and SGLT2is across CKD stages, and on referral criteria for nephrology co-management based on kidney function, proteinuria, and clinical red flags.
Recommendations 9–10 were developed with reference to the KDIGO 2024 CKD Guideline and the 2025 Taiwan CKD Clinical Practice Guideline, with thresholds aligned to the Taiwan CKD Pay-for-Performance Programs.
Abbreviations list: ACEis/ARBs: Angiotensin-Converting Enzyme Inhibitors/Angiotensin II Receptor Blockers; SGLT2i: Sodium-Glucose Cotransporter-2 Inhibitor; uACR: Urine Albumin-to-Creatinine Ratio; eGFR: Estimated Glomerular Filtration Rate; uPCR: Urine Protein-to-Creatinine Ratio.

and should be prioritized as foundational therapies in CKD. They are generally continued until dialysis or transplantation unless contraindicated.

Other Emerging Therapies: Because current large trials of non-steroidal mineralocorticoid receptor antagonists (FIDELIO, FIGARO, and FIDELITY studies) and Glucagon-like peptide-1 receptor agonists (GLP-1 RAs; e.g., semaglutide in the FLOW and SELECT studies) have primarily enrolled patients with type 2 diabetes with CKD [59–63]. Consequently, evidence for their use in non-diabetic CKD remains limited. In Taiwan, reimbursement and accessibility constraints further restrict their applicability to the broader CKD population. Nevertheless, both drug classes provide high-quality evidence of cardiorenal benefit in diabetic CKD and show strong potential to become future cornerstone therapies once validated in non-diabetic CKD. Their omission from the formal recommendations does not imply discouragement of use in eligible patients with diabetes, obesity, or established cardiovascular disease.

3.10. Recommendation 10 - Rationale

This recommendation aims to ensure timely nephrology involvement for CKD patients at high risk of progression or with atypical clinical courses. Early nephrology consultation or referral confirms diagnosis, identifies reversible causes, optimizes treatment, and prepares for advanced CKD care. It is associated with smoother care transitions and fewer unplanned dialysis starts [21,24,64–66].

Among these criteria, declining eGFR and severe proteinuria indicate disease severity, while abrupt changes may signal reversible causes such as dehydration, drug-induced injury, obstruction, or new-onset glomerular disease [67,68]. Persistent hematuria, unexplained electrolyte disorders, or resistant hypertension suggest atypical or secondary processes requiring further evaluation, including glomerulonephritis, urologic, endocrine, or renovascular disease. Abrupt kidney function decline in a diabetic patient without retinopathy should also raise concern for alternative diagnoses.

Building on these considerations, our referral criteria share common ground with KDIGO 2024 CKD Guideline and the 2025 Taiwan Clinical Practice Guideline for CKD in stressing timely nephrology involvement for patients with reduced kidney function, progressive decline, or significant proteinuria. All three highlight eGFR, proteinuria, and rapid deterioration as key triggers, but differ in approach. KDIGO 2024 CKD Guideline adopts a risk-based strategy, recommending referral when 5-year kidney failure risk exceeds 3–5 %, whereas the 2025 Taiwan guideline relies on earlier eGFR- and albuminuria-based thresholds practical for frontline care. Our consensus aligns more with Taiwan practice, specifying eGFR <45 mL/min/1.73 m^2, rapid decline >5 mL/min/1.73 m^2/year, severe proteinuria, and clinical red flags such as resistant hypertension, refractory electrolyte disorders, or suspected systemic, hereditary, or structural disease. We did not adopt a Kidney Failure Risk Equation cut-off, given limited local validation and the need for clear, actionable criteria [21,24,25,69]. The eGFR <45 mL/min/1.73 m^2 threshold is consensus-based, aligned with the Pre-ESKD program's enrollment criteria to enable timely referral into structured multidisciplinary care.

In Taiwan, "consultation or referral" reflects the nephrologist's role as a co-managing partner in the Early-CKD P4P program, while primary care physicians remain central in managing comorbidities. Unlike international guidelines, which generally recommend referral at eGFR <30 mL/min/1.73 m^2 or based on risk equations, Taiwan uses a threshold of <45 mL/min/1.73 m^2 to facilitate earlier enrollment into the Pre-ESKD P4P program [33]. This consensus therefore emphasizes pragmatic, earlier triggers tailored to frontline physicians, contrasting with the more specialist-oriented 2025 Taiwan Clinical Practice Guideline for CKD and the later-stage, risk-based approach of KDIGO 2024 CKD Guideline [21,24].

Taken together, Recommendation 10 provides a proactive and practical referral framework that balances international evidence with Taiwan's health-care priorities, aiming to enhance care transitions and reduce the future dialysis burden.

4. Conclusion

Effective CKD care in Taiwan requires practical strategies in primary care and timely nephrology involvement. This consensus highlights risk identification, simple diagnostic tools, clear staging and referral criteria, individualized targets for blood pressure, glycemia, and lipids, as well as health literacy, lifestyle modification, and cornerstone pharmacotherapies. Integrated into Taiwan's P4P programs, these recommendations support coordinated primary care, early nephrology referral, and reduced CKD progression, dialysis burden, and mortality.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used ChatGPT in order to edit English grammar. After using this tool/service, the author (s) reviewed and edited the content as needed and take full responsibility for the content of the published article.

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